EXECUTIVE ACTION PLAN

TOP 12 STRATEGIC RECOMMENDATIONS

UK National Railway Performance, Revenue, and Operations Intelligence

A stakeholder-focused strategic roadmap synthesizing commercial value, operational reliability, customer impact, route economics, time-of-day risk, ticket-class performance, delay severity, and refund exposure.

STRATEGIC PRIORITY	Protect the network's highest-value revenue pools, resolve structural operational vulnerabilities, strengthen the premium proposition, reduce customer-impact leakage, and institutionalize evidence-based executive prioritization.

Executive Strategic Overview

The twelve recommendations are organized around four strategic pillars. Together, they move the dashboard from performance reporting toward a management system for protecting value, reducing structural risk, strengthening customer outcomes, and prioritizing investment.

01 PROTECT CRITICAL REVENUE	**Recommendations 1–3**	Secure the highest-value operating window and the two most commercially exposed high-yield corridors.
02 FIX STRUCTURAL OPERATIONAL RISK	**Recommendations 4–7**	Target disruption by time, cause, severity, operating context, and demonstrated internal benchmarks.
03 STRENGTHEN CUSTOMER & PREMIUM VALUE	**Recommendations 8, 9 & 11**	Protect premium economics, replicate sustainable excellence, and reduce commercial leakage from disruption.
04 INSTITUTIONALIZE SMARTER DECISION-MAKING	**Recommendations 10 & 12**	Transform route management and executive prioritization through strategic archetypes and integrated value-at-risk scoring.

Where management should act first

Priority 1 — Morning Peak Reliability. Protect the network's largest recurring revenue pool.
Priority 2 — Liverpool Lime Street → London Euston. Resolve the clearest large-scale Value-at-Risk corridor.
Priority 3 — Manchester Piccadilly → London Euston. Protect exceptionally high-yield journeys from persistent reliability failure.

01 Make Morning Peak Reliability the Network's No. 1 Revenue-Protection Priority

BUSINESS CASE	Morning Peak is simultaneously the network's largest economic engine and its greatest operating vulnerability. It generates approximately **£301.65K, or 40.7% of total network revenue**, with the highest revenue per trip at **£32.38**. Yet it also records the **lowest on-time rate at 89%, highest delay rate at 11.07%, and highest average delay at 47.23 minutes**. This is the clearest concentration of commercial value at operational risk anywhere in the network. Morning Peak generates **89% more revenue than Off-Peak**, yet its delay rate is more than four times higher.
RECOMMENDED ACTION	Establish a dedicated **Morning Peak Reliability Programme** focused on the highest-value routes and dominant disruption causes. Protect peak-period capacity, strengthen operational readiness before the morning service window, position contingency resources around critical corridors, and prioritize preventive action on recurring weather, signal, and route-specific vulnerabilities.
EXPECTED BUSINESS IMPACT	Protection of the network's largest revenue pool, fewer passenger disruptions during its most commercially important window, reduced refund exposure, and stronger overall network reliability.

02 Launch an Immediate Turnaround Programme for Liverpool Lime Street → London Euston

BUSINESS CASE	This is the network's clearest large-scale **Value-at-Risk corridor**. It consistently ranks second in revenue, generates exceptional revenue per trip, yet suffers chronically poor punctuality. The evidence is particularly severe: • Monthly on-time performance remains only **20%–36%**. • During Morning Peak, the route generates approximately **£91K—around 30% of all Morning Peak revenue—with £110 revenue per trip, but only 22% on-time performance**. • Standard Class alone generates approximately **£99K at £99 per trip**, with only **29% punctuality and 158 refunds**. • First Class reaches an exceptional **£146 per trip**, yet punctuality remains only **28%**. This is not an isolated time-period or ticket-class issue. The reliability weakness is structural.
RECOMMENDED ACTION	Create a route-specific turnaround programme with clear executive ownership. Conduct root-cause decomposition by time of day and delay cause, establish punctuality recovery targets, identify recurring infrastructure and timetable constraints, and monitor refunds alongside operational KPIs.
EXPECTED BUSINESS IMPACT	Direct protection of a major revenue corridor, reduction in refund leakage, improved experience for high-value passengers, and lower strategic dependency risk around one of the network's most economically important routes.

03 Protect High-Yield Journeys by Fixing the Manchester Piccadilly → London Euston Reliability Contradiction

BUSINESS CASE	Manchester Piccadilly → London Euston presents the network's most extreme **yield-versus-reliability contradiction**. It generates exceptional value per journey while delivering consistently poor punctuality. Revenue per trip remains approximately **£88–£110 across months**, while on-time performance stays between only **26% and 39%**. The contradiction becomes even more pronounced by ticket class: First Class generates **£153 per trip—the highest visible premium yield—yet only 22% of journeys are on time**. Time-of-day performance is equally concerning, with punctuality falling to just **15% during Morning Peak and 4% during Evening Peak**.
RECOMMENDED ACTION	Treat this corridor as a dedicated **high-yield value-protection case**, distinct from high-volume operational problems. Identify why performance remains weak across virtually every time window and both ticket classes. Prioritize timetable resilience, bottleneck diagnosis, delay propagation analysis, and proactive communication for affected premium passengers.
EXPECTED BUSINESS IMPACT	Protection of some of the network's highest-value individual journeys, stronger premium-service credibility, and reduced risk of losing commercially valuable demand because of persistently poor reliability.

04 Build Time-Specific Delay Strategies Instead of Using One Network-Wide Intervention Model

BUSINESS CASE	The dominant cause and nature of operational failure change fundamentally throughout the day: • **Morning Peak:** Weather dominates with **829 incidents**. • **Midday:** Staff shortage leads with **303 incidents**. • **Evening Peak:** Signal failure dominates with **397 incidents**. • **Off-Peak:** The network achieves **97% punctuality and only a 2.64% delay rate**. The failure mode also changes. Morning Peak has the highest delay frequency, Midday shows greater delay-escalation and refund exposure, and Evening Peak has the highest reported Severe Delay Share at **25%**. A generic network-wide delay-reduction programme would therefore allocate resources inefficiently because the same intervention cannot solve weather exposure, staffing constraints, signal failures, and severe-delay escalation equally well.
RECOMMENDED ACTION	Introduce a **Time-of-Day Operational Risk Playbook** with differentiated intervention strategies: weather preparedness and service resilience for Morning Peak, workforce and rostering resilience for Midday, signal reliability and infrastructure response for Evening Peak, and benchmarking of Off-Peak operating practices.
EXPECTED BUSINESS IMPACT	Better allocation of operational resources, faster targeting of root causes, fewer recurring incidents, and stronger returns on reliability investment.

05 Turn Off-Peak Performance into the Internal Reliability Benchmark for the Entire Network

BUSINESS CASE	Off-Peak demonstrates that commercial value and operational reliability can coexist. It generates approximately **£159.42K—21.5% of total network revenue—while achieving 97% on-time performance, only a 2.64% delay rate, £20.42 revenue per trip, and the lowest refund rate at 2.19%**. The most significant finding is the consistency of its reliability. Across every month, Off-Peak punctuality remains approximately **97%**, while Morning Peak varies between only 87% and 91%. This suggests that the network possesses the operational capability to achieve very high reliability, but that capability deteriorates under specific capacity, demand, infrastructure, or complexity conditions.
RECOMMENDED ACTION	Conduct a formal **Peak-versus-Off-Peak operational gap analysis**. Compare turnaround margins, capacity pressure, timetable density, staffing coverage, infrastructure utilization, delay propagation, and recovery flexibility. Identify which Off-Peak practices can realistically be transferred to peak operations.
EXPECTED BUSINESS IMPACT	A data-driven internal benchmark for reliability improvement, more realistic operational targets, and reduced dependence on external standards that may not reflect the network's own demonstrated capability.

06 Redesign Midday as a Combined Efficiency and Customer-Impact Turnaround Opportunity

BUSINESS CASE	Midday is the weakest overall business window. It generates the **lowest revenue at £101.59K, lowest trip volume at approximately 6K, and lowest revenue per trip at £17.54**, yet simultaneously records: • **Highest refund rate: 7.03%** • **Highest total refunds: 407** • **Critical Delay Threshold: 161** • **Delay rate: 7.86%** • Significant severe-delay exposure This means Midday absorbs substantial operational failure and customer-impact costs without generating proportionate commercial value. The most striking contradiction is that Midday produces roughly one-third of Morning Peak revenue but almost twice as many refunds—**407 versus 209**.
RECOMMENDED ACTION	Create a dedicated **Midday Efficiency Review** covering routes, staffing, delay escalation, refund-generating journeys, and capacity utilization. Give particular attention to Manchester Piccadilly → Liverpool Lime Street, where Midday average delay reaches an extraordinary **109 minutes** despite revenue per trip of only around **£4**.
EXPECTED BUSINESS IMPACT	Lower refund exposure, better resource productivity, reduced severe-delay escalation, and improved economics in the network's weakest operating period.

07 Separate Delay Prevention from Severe-Delay Containment

BUSINESS CASE	The insight bank proves that **delay frequency and delay severity are different operational problems**. March had the highest delay rate at **7.81%** but only a **12% Severe Delay Share**, whereas January and April had lower delay rates but a higher **17% Severe Delay Share**. The same pattern appears across the operating day: • Morning Peak: **11.07% delay rate**, but only **10% Severe Delay Share**. • Midday: Lower **7.86% delay rate**, but **16% Severe Delay Share**. • Evening Peak: **6.66% delay rate**, yet **25% Severe Delay Share**. A single average-delay KPI therefore cannot distinguish between frequent small failures and fewer incidents that escalate catastrophically.
RECOMMENDED ACTION	Establish two separate operational workstreams: **Prevention:** Reduce how frequently delays begin through maintenance, weather readiness, staffing resilience, signalling reliability, and schedule design. **Containment:** Prevent existing disruptions from escalating through rapid-response protocols, service recovery, rerouting, passenger management, and escalation thresholds.
EXPECTED BUSINESS IMPACT	More precise operational management, fewer severe incidents, faster recovery from disruption, and better alignment between interventions and actual failure mechanisms.

08 Restore the Premium Value Proposition by Closing the First Class Reliability Gap

BUSINESS CASE	First Class is the network's **yield engine**. It represents less than **10% of trips but approximately 20% of revenue**, generating **£48.86 per trip versus £20.72 for Standard—a 136% yield premium**. Yet First Class receives no corresponding reliability premium. In fact, every major reliability indicator is slightly worse: • On-time rate: **92% vs 93%** • Delay rate: **7.65% vs 7.13%** • Severe Delay Share: **17% vs 14%** • Refund rate: **3.70% vs 3.51%** Premium passengers therefore create dramatically more economic value per journey without receiving measurably better operational reliability.
RECOMMENDED ACTION	Introduce a **Premium Value Protection framework**. Track First Class performance explicitly on high-yield corridors, prioritize severe-delay reduction where premium revenue is concentrated, and strengthen disruption recovery and passenger communication when reliability cannot be immediately restored. The objective should not be to create an entirely separate railway for First Class, but to ensure that the premium proposition is not undermined by persistent operational weakness.
EXPECTED BUSINESS IMPACT	Better protection of high-yield revenue, stronger premium-service credibility, reduced refund and retention risk, and improved alignment between price premium and delivered experience.

09 Replicate the London Kings Cross → York Network Champion Model

BUSINESS CASE	London Kings Cross → York is the strongest example of sustainable commercial and operational performance in the dataset. It remains the top revenue route in every month while maintaining approximately **96%–98% punctuality and relatively contained average delays of 13–18 minutes**. Its strength persists across every time window: • Morning Peak: **£63K revenue, 95% on-time** • Midday: **£23K, 100% on-time** • Evening Peak: **£57K, 95% on-time** • Off-Peak: **£40K, 100% on-time** It also performs strongly in both Standard and First Class, demonstrating that high revenue, premium monetization, and strong reliability can coexist.
RECOMMENDED ACTION	Use London Kings Cross → York as the network's primary **Network Champion benchmark**. Study its timetable structure, operational processes, route characteristics, recovery mechanisms, passenger mix, and resource deployment. Identify transferable practices for other commercially important corridors. The purpose is not to assume every route can replicate identical performance, but to determine what management can learn from the network's strongest proven example.
EXPECTED BUSINESS IMPACT	Evidence-based operational improvement, faster adoption of proven internal practices, and a more practical benchmark for sustainable revenue and reliability performance.

10 Manage Routes Through Strategic Archetypes Rather Than One Universal KPI Ranking

BUSINESS CASE	The network contains fundamentally different models of value creation and risk. Ranking all routes by revenue, trips, or punctuality alone hides these differences. The evidence supports at least four strategic route archetypes: **Network Champions:** High commercial value and strong reliability—exemplified by London Kings Cross → York. **Value at Risk:** High commercial value but poor reliability—exemplified by Liverpool Lime Street → London Euston. **High-Yield Vulnerabilities:** Lower-volume journeys with exceptional monetization but weak service quality—exemplified by Manchester Piccadilly → London Euston. **High-Volume Operational Burdens:** Large passenger exposure, low yield, and significant disruption—exemplified by Manchester Piccadilly → Liverpool Lime Street. Each archetype requires a different management response. A high-volume, £4-per-trip corridor should not be managed using the same logic as a £153-per-trip premium corridor.
RECOMMENDED ACTION	Introduce a **Route Prioritization Matrix** based on commercial value, passenger volume, revenue per trip, punctuality, delay severity, and refund exposure. Assign differentiated actions to each route archetype: protect, fix urgently, grow selectively, optimize efficiency, or strategically review.
EXPECTED BUSINESS IMPACT	Better capital and management-resource allocation, clearer executive prioritization, and reduced risk of treating economically different routes as operationally equivalent.

11 Use Refunds as an Early-Warning Signal Connecting Operational Failure to Commercial Leakage

BUSINESS CASE	Refunds provide the project's clearest bridge between service disruption and measurable customer impact: **Operational disruption → Passenger inconvenience → Refund exposure → Revenue leakage** However, refund behavior does not simply follow revenue or delay frequency. Midday has a **7.03% refund rate and 407 refunds**, despite generating the lowest revenue. Morning Peak has the highest delay rate but only a **2.24% refund rate**. March simultaneously combines strong revenue with the highest refund rate and refund count. These contradictions indicate that refund exposure is driven by the interaction of route, time period, passenger mix, ticket conditions, and disruption severity.
RECOMMENDED ACTION	Establish a **Refund Risk Early-Warning System** that monitors refund rates alongside route value, delay severity, time of day, and ticket class. Prioritize cases where refund exposure rises disproportionately relative to delay frequency or revenue contribution. Refunds should not be viewed merely as an accounting outcome after disruption. They should be used diagnostically to identify where operational failures are producing the greatest commercial and customer consequences.
EXPECTED BUSINESS IMPACT	Earlier detection of customer-impact hotspots, reduced revenue leakage, better prioritization of service recovery, and stronger connection between operational and commercial decision-making.

12 Build an Integrated Value-at-Risk Decision Engine for Executive Prioritization

BUSINESS CASE	The project's greatest strategic value is not simply reporting what happened. It is identifying **where management should act first**. No individual KPI is sufficient. Revenue alone ignores reliability. Delay rate ignores commercial importance. Average delay ignores passenger scale. Refunds alone do not explain the operational cause. The strongest intervention priorities emerge where multiple forms of exposure overlap: **Economic importance + Passenger exposure + Operational weakness + Delay severity + Refund risk** This framework consistently identifies Morning Peak, Liverpool Lime Street → London Euston, Manchester Piccadilly → London Euston, Midday refund exposure, and specific high-volume corridors as strategically important—but for different reasons.
RECOMMENDED ACTION	Develop an executive **Value-at-Risk Prioritization Score** using a carefully weighted combination of: **Revenue exposure × Trip volume × Revenue per trip × Reliability gap × Delay severity × Refund exposure** The score should not replace management judgment. It should create a consistent evidence-based mechanism for deciding which routes, time periods, and service segments warrant immediate intervention, protection, investment, or review.
EXPECTED BUSINESS IMPACT	Faster executive decision-making, more disciplined investment prioritization, stronger alignment between commercial and operational teams, and better concentration of resources where intervention can protect or create the greatest business value.